

# MA388 Sabermetrics: Lesson 26

## Statist Spray Charts

```
library(tidyverse)
library(knitr)
library(baseballr)
```

Review

**Last class we talked about the streakiness or clumpiness coefficient. What does this metric measure and how do we calculate it?**

**True or False. If a player had a high batting average for the season, this means that they were not a streaky player.**

**Could a player be temporally streaky but spatially predictable?**

What do Joey Gallo, Albert Pujols, and DJ LeMahieu have in common?

*Here's one answer:* They had some of the most extreme spray charts among MLB hitters in 2017.

Of course, all three hitters went to different extremes – Gallo was one of MLB’s most pull-heavy hitters, Pujols hit balls to center field more than almost anyone, and LeMahieu used the opposite field like nobody else in baseball.

Here is the full article: [Extreme pull and opposite-field hitters \(MLB.com, 2017\)](#)

Let’s explore their spray charts for that year.

```
library(tidyverse)
library(baseballr)

gallo <- scrape_statcast_savant(start_date = "2017-05-01",
                                end_date = "2017-12-31",
                                playerid = 608336,
                                player_type = "batter")

pujols <- scrape_statcast_savant(start_date = "2017-05-01",
                                  end_date = "2017-12-31",
                                  playerid = 405395,
                                  player_type = "batter")

lemahieu <- scrape_statcast_savant(start_date = "2017-05-01",
                                    end_date = "2017-12-31",
                                    playerid = 518934,
                                    player_type = "batter")

all_data = rbind(gallo, lemahieu, pujols)
```

Filter the data to only include balls in play.

```
all_bip <- all_data |>
  filter(type == "X")

# hc_x and hc_y are the hit coordinates for balls in play.
all_bip |>
  select(player_name, events, hc_x, hc_y) |>
  head()
```

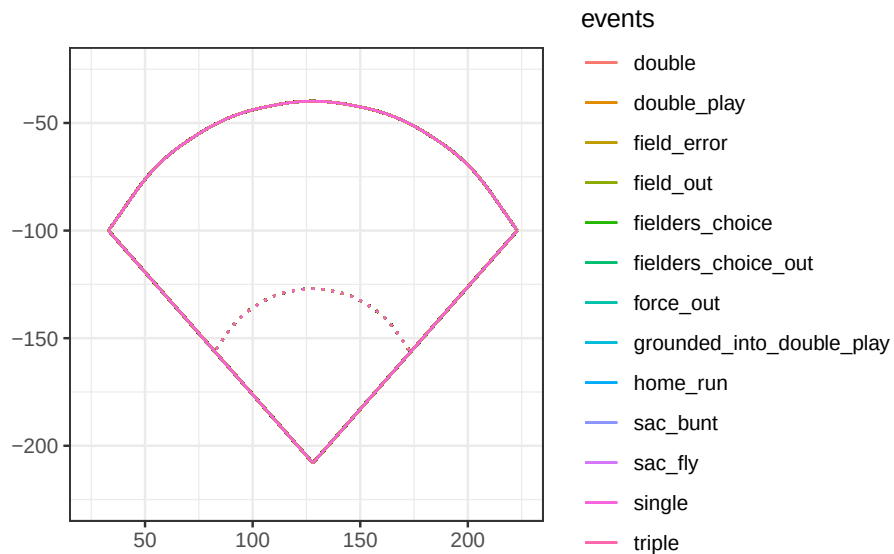
|    | player_name | events    | hc_x   | hc_y   |
|----|-------------|-----------|--------|--------|
|    | <char>      | <char>    | <num>  | <num>  |
| 1: | Gallo, Joey | home_run  | 115.47 | 14.79  |
| 2: | Gallo, Joey | home_run  | 227.17 | 67.05  |
| 3: | Gallo, Joey | field_out | 156.70 | 154.25 |
| 4: | Gallo, Joey | single    | 126.94 | 85.99  |
| 5: | Gallo, Joey | field_out | 154.54 | 74.44  |
| 6: | Gallo, Joey | field_out | 170.77 | 144.09 |

Now let's create the base plot for our spray chart.

```
spray_chart <- function(...) {
  ggplot(...) +
    geom_curve(x = 33, xend = 223, y = -100, yend = -100,
              curvature = -.65) +
    geom_segment(x = 128, xend = 33, y = -208, yend = -100) +
    geom_segment(x = 128, xend = 223, y = -208, yend = -100) +
    geom_curve(x = 83, xend = 173, y = -155, yend = -155,
              curvature = -.65, linetype = "dotted") +
    coord_fixed() +
    scale_x_continuous(NULL, limits = c(25, 225)) +
    scale_y_continuous(NULL, limits = c(-225, -25))
}
```

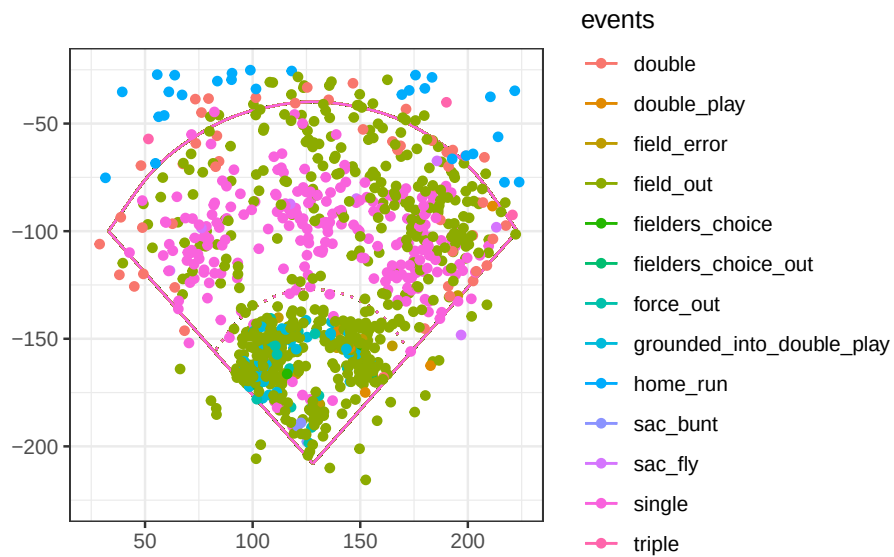
Where would we expect these different events to occur?

```
spray_chart(all_bip, aes(x = hc_x, y = -hc_y, color = events)) +
  scale_color_hue()+
  theme_bw()
```



Let's look at all three players together.

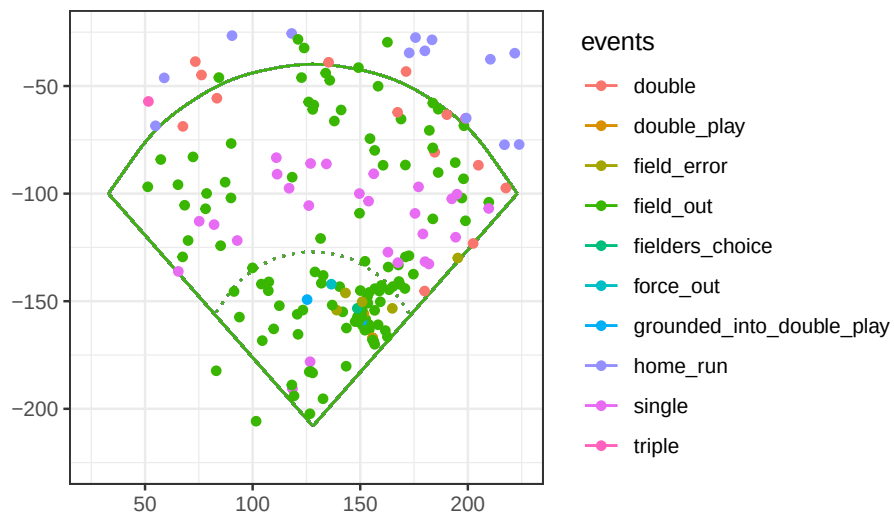
```
spray_chart(all_bip, aes(x = hc_x, y = -hc_y, color = events)) +
  geom_point() +
  scale_color_hue()+
  theme_bw()
```



Now let's take a look at each player individually.

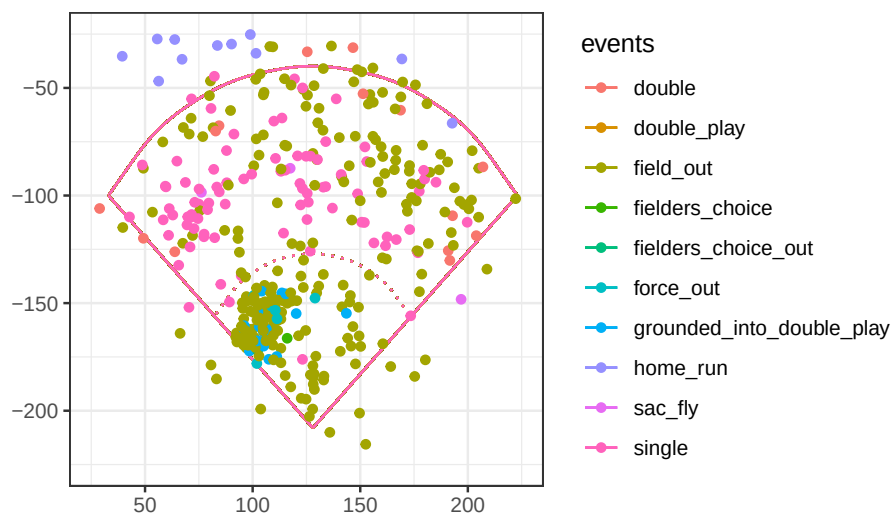
```
all_bip |>
  filter(player_name=="Gallo, Joey") |>
  spray_chart(aes(x = hc_x, y = -hc_y, color = events)) +
  geom_point() +
  scale_color_hue()+
  labs(title = "Spray Chart Joey Gallo (Left Handed)") +
  theme_bw()
```

Spray Chart Joey Gallo (Left Handed)

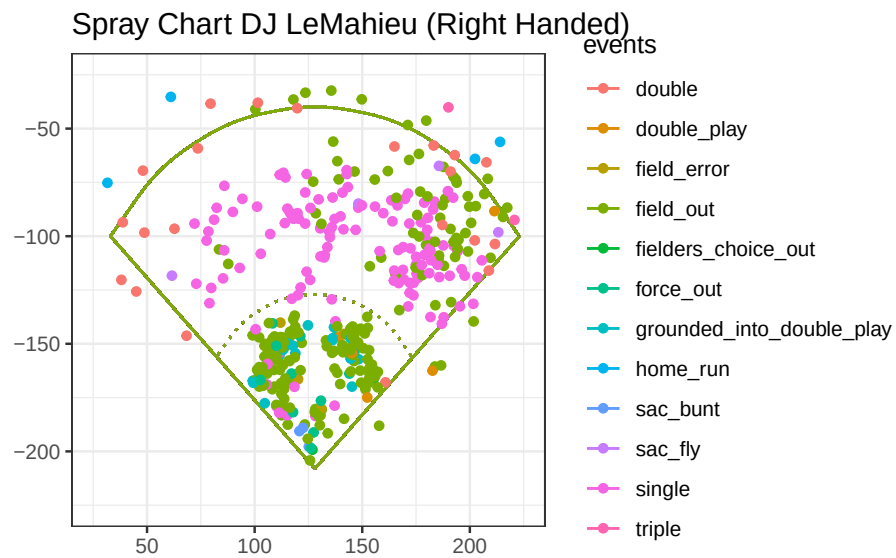


```
all_bip |>
  filter(player_name=="Pujols, Albert") |>
  spray_chart(aes(x = hc_x, y = -hc_y, color = events)) +
  geom_point() +
  scale_color_hue()+
  labs(title = "Spray Chart Albert Pujols (Right Handed)") +
  theme_bw()
```

Spray Chart Albert Pujols (Right Handed)

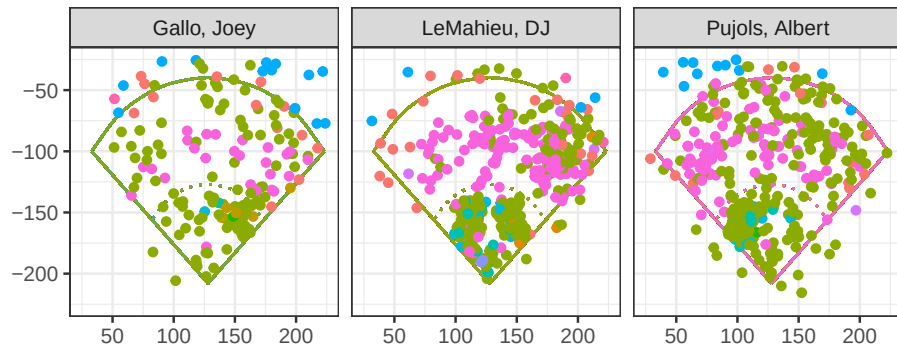


```
all_bip |>
  filter(player_name=="LeMahieu, DJ") |>
  spray_chart(aes(x = hc_x, y = -hc_y, color = events)) +
  geom_point() +
  scale_color_hue()+
  labs(title = "Spray Chart DJ LeMahieu (Right Handed)")+
  theme_bw()
```



For a side-by-side comparison:

```
all_bip |>
  spray_chart(aes(x = hc_x, y = -hc_y, color = events)) +
  geom_point() +
  scale_color_hue()+
  theme_bw()+
  theme(legend.position = "none")+
  facet_wrap(~player_name)
```



```
# Traditional defensive alignment
traditional_defense <- tibble(
  position = c("P", "C", "1B", "2B", "SS", "3B", "LF", "CF", "RF"),
  x = c(128, 128, 160, 140, 115, 95, 80, 128, 180),
  y = c(-165, -220, -150, -135, -135, -150, -80, -60, -80)
)

# Shift vs pull-heavy LEFTY (e.g., Gallo)
shift_defense <- tibble(
  position = c("P", "C", "1B", "2B_shift", "SS_shift", "3B_shift", "LF", "CF", "RF"),
  x = c(128, 128, 165, 155, 135, 110, 90, 135, 185),
  y = c(-165, -220, -150, -115, -125, -135, -85, -65, -85)
)

add_defense <- function(defense_data, color = "black") {
  list(
    geom_point(data = defense_data,
              aes(x = x, y = y),
              color = color,
              size = 3),

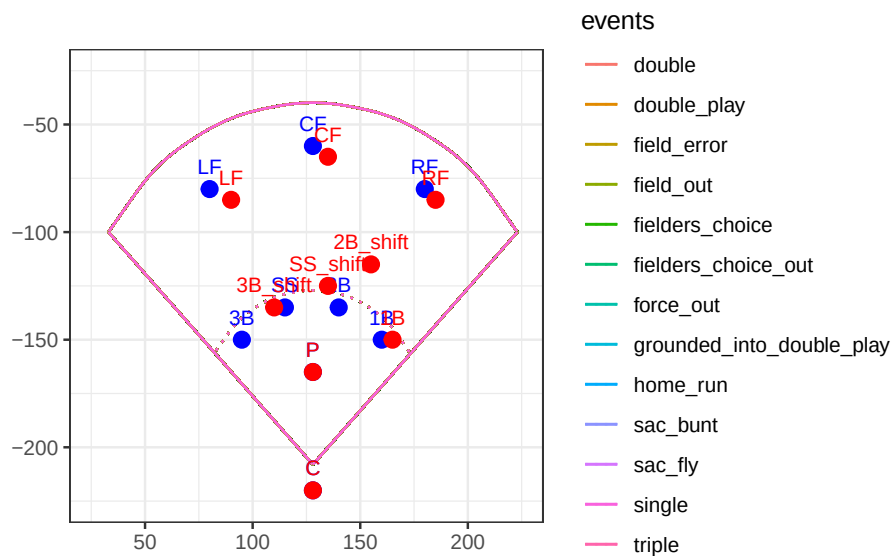
    geom_text(data = defense_data,
             aes(x = x, y = y, label = position),
             vjust = -1,
             color = color,
             size = 3)
  )
}
```

```

    )
  }

all_bip |>
  # filter(player_name == "Gallo, Joey") |>
  spray_chart(aes(x = hc_x, y = -hc_y, color = events)) +
  # geom_point(alpha = 0.6) +
  add_defense(traditional_defense, color = "blue") +
  add_defense(shift_defense, color = "red") +
  # labs(title = "Joey Gallo vs Traditional Defense") +
  theme_bw()

```



In general, players hit ground balls to the pull side of the field. Let's take a look at how these three players hit their ground balls and the percentage of times they faced infield shifts.

```

left_handed_batters = gallo |>
  filter(launch_angle < 10) |>
  summarize(N = n(),
    Pulled = 100 * mean(hc_x > 125.42, na.rm = TRUE),
    Shift = 100 *
      mean(if_fielding_alignment == "Infield shift",
        na.rm = TRUE), .by='player_name')

right_handed_batters = all_data |>
  filter(player_name!="Gallo, Joey") |>

```



```

filter(launch_angle < 10) |>
summarize(N = n(),
          Pulled = 100 * mean(hc_x < 125.42, na.rm = TRUE),
          Shift = 100 *
            mean(if_fielding_alignment == "Infield shift",
                 na.rm = TRUE), .by='player_name')

rbind(left_handed_batters, right_handed_batters)

```

```

# A tibble: 3 x 4
  player_name      N Pulled Shift
  <chr>      <int> <dbl> <dbl>
1 Gallo, Joey      93   91.5   87.1
2 LeMahieu, DJ    312   49.8    0
3 Pujols, Albert  240   83.0   40

```

What does this tell us about the tendencies of these batters and how defenses approached them?